

# KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness enables autonomous regulation of acting, reflection, and knowledge retrieval.

Source: [arxiv.org/abs/2504.03553](https://arxiv.org/abs/2504.03553)

# Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge

## Three dominant paradigms:

- Trajectory imitation (ReAct): pattern-following from demonstrations
- Trial-and-error refinement (Reflexion): iterative self-feedback
- Knowledge-augmented planning (KnowAgent/WKM): always-on retrieval

Common limitation: injected signals are applied regardless of step-level need.

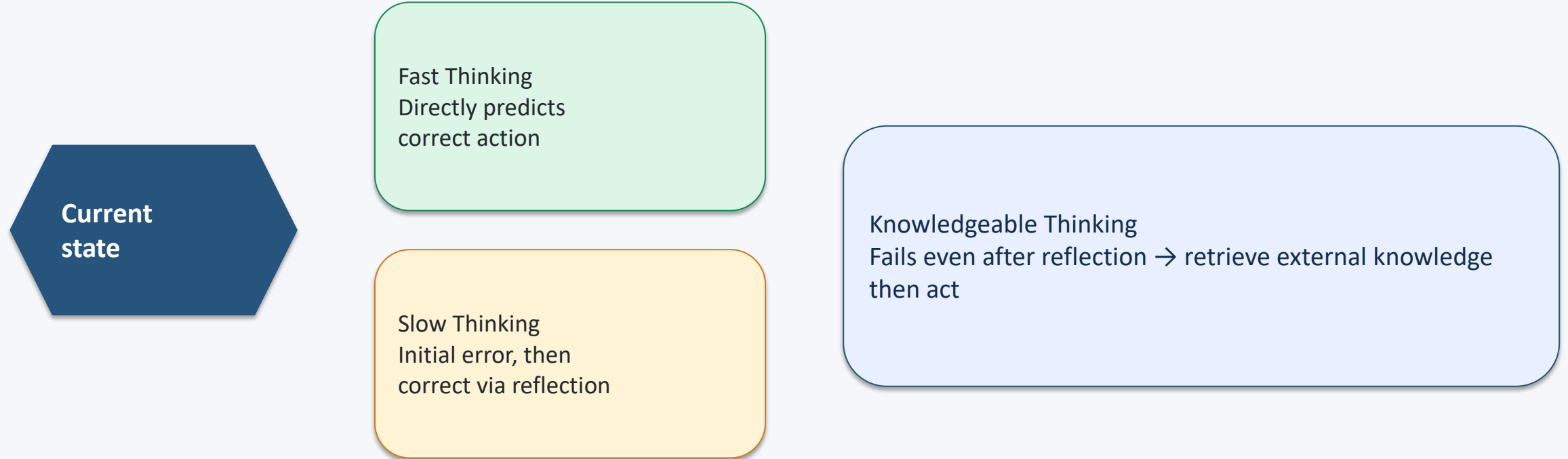
Consequences: fragile planning, pattern collapse on novel states, and inflated inference cost.

## Human-style situational self-awareness

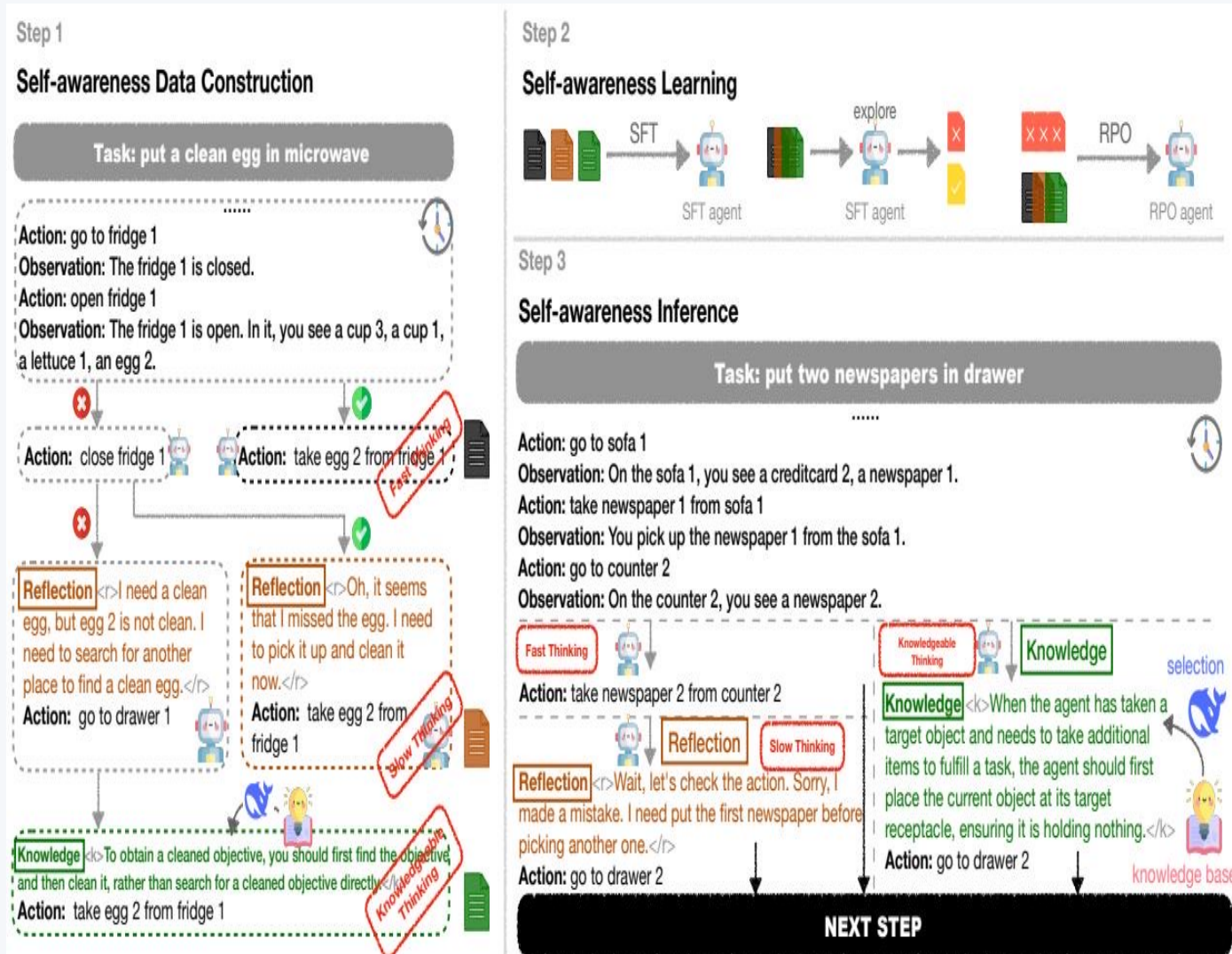
- 1) Can I act directly?
- 2) Should I pause and reflect?
- 3) Do I need external knowledge?

KnowSelf operationalizes this metacognitive control for LLM agents.

# Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable



# KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aw



## 1) Data construction

Heuristic situation judgment labels self-explored trajectories with special tokens for fast/slow/knowledgeable states.

## 2) Two-stage training

SFT initializes self-awareness patterns; RPO refines policy to emit appropriate control tokens.

## 3) Self-aware inference

Generated tokens dynamically trigger direct action, reflection, or retrieval at runtime.

# KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines

| (a) Task: put a hot mug in cabinet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>.....</p> <p><b>Action:</b> take mug 1 from countertop 1</p> <p><b>Observation:</b> You pick up the mug 1 from the countertop 1.</p> <p><b>Action:</b> go to microwave 1</p> <p><b>Observation:</b> The microwave 1 is closed.</p> <p><b>Action:</b> open microwave 1</p> <p><b>Observation:</b> You open the microwave 1. In it, you see an egg 1.</p> <p><b>[Knowledge]</b>&lt;k&gt;If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.&lt;/k&gt;</p> <p><b>Action:</b> take egg 1 from microwave 1</p> <p><b>Observation:</b> Nothing happens.</p> | <p>.....</p> <p><b>Action:</b> take mug 1 from countertop 1</p> <p><b>Observation:</b> You pick up the mug 1 from the countertop 1.</p> <p><b>Action:</b> go to microwave 1</p> <p><b>Observation:</b> The microwave 1 is closed.</p> <p><b>Action:</b> open microwave 1</p> <p><b>Observation:</b> You open the microwave 1. In it, you see an egg 1.</p> <p><b>Action:</b> put mug 1 in/on microwave 1</p> <p><b>[Reflection]</b>&lt;r&gt;Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.&lt;/r&gt;</p> <p><b>Action:</b> heat mug 1 with microwave 1</p> <p>.....</p>                    |
| (b) Task: put a saltshaker in drawer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>.....</p> <p><b>Action:</b> take saltshaker 1 from countertop 3</p> <p><b>Observation:</b> You pick up the saltshaker 1 from the countertop 3.</p> <p><b>Action:</b> go to drawer 1</p> <p><b>Observation:</b> The drawer 1 is open. In it, you see a fork 1.</p> <p><b>Action:</b> put saltshaker 1 in/on drawer 1</p> <p><b>[Reflection]</b>&lt;r&gt;Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.&lt;/r&gt;</p> <p><b>Action:</b> close drawer 1</p> <p><b>Observation:</b> You close the drawer 1.</p> <p><b>Action:</b> put saltshaker 1 in/on drawer 1</p> <p><b>Observation:</b> Nothing happens.</p>                                                                                   | <p>.....</p> <p><b>Action:</b> take saltshaker 1 from countertop 3</p> <p><b>Observation:</b> You pick up the saltshaker 1 from the countertop 3.</p> <p><b>Action:</b> go to drawer 1</p> <p><b>Observation:</b> The drawer 1 is closed.</p> <p><b>Action:</b> open drawer 1</p> <p><b>Observation:</b> You open the drawer 1. In it, you see a fork 1.</p> <p><b>[Knowledge]</b>&lt;k&gt;When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.&lt;/k&gt;</p> <p><b>Action:</b> put saltshaker 1 in/on drawer 1</p> |

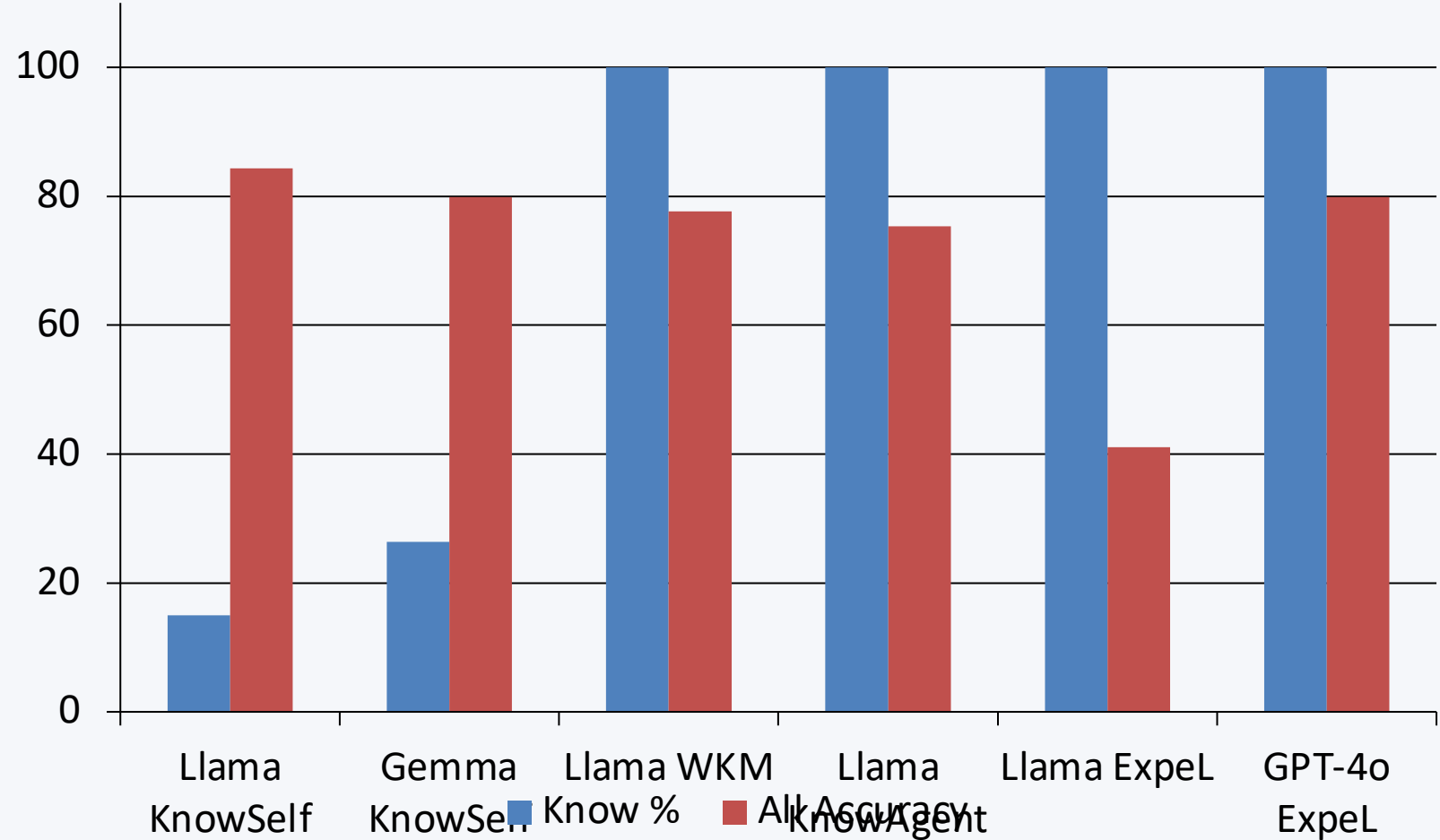
Left: 'put a hot mug in cabinet'—KnowSelf uses reflection to heat mug. Right: 'put a saltshaker in drawer'—KnowSelf invokes knowledge to avoid spurious drawer-closing b



# KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

| Backbone | Method    | Know%  | Put    | Clean | Heat  | Cool  | Examine | Put Two | All   |
|----------|-----------|--------|--------|-------|-------|-------|---------|---------|-------|
| GPT-4o   | ReAct     | 0%     | 83.33  | 74.19 | 69.57 | 85.71 | 77.78   | 64.71   | 76.12 |
| GPT-4o   | Reflexion | 0%     | 100.00 | 87.10 | 73.91 | 90.48 | 83.33   | 70.59   | 85.07 |
| GPT-4o   | ExpeL     | 100%   | 95.83  | 83.87 | 69.57 | 80.95 | 88.89   | 52.94   | 79.85 |
| Llama-8B | ReAct     | 0%     | 33.33  | 3.23  | 0.00  | 57.14 | 66.67   | 23.53   | 27.61 |
| Llama-8B | Reflexion | 0%     | 62.96  | 22.73 | 5.88  | 64.29 | 86.36   | 50.00   | 51.49 |
| Llama-8B | ExpeL     | 100%   | 83.33  | 32.26 | 30.43 | 23.81 | 55.56   | 17.65   | 41.04 |
| Llama-8B | ETO       | 0%     | 91.67  | 70.59 | 82.61 | 61.90 | 88.89   | 64.71   | 78.36 |
| Llama-8B | KnowAgent | 100%   | 87.50  | 93.55 | 65.22 | 66.67 | 61.11   | 64.71   | 75.37 |
| Llama-8B | WKM       | 100%   | 95.83  | 87.10 | 86.96 | 61.90 | 66.67   | 52.94   | 77.61 |
| Llama-8B | KnowSelf  | 15.01% | 91.67  | 87.10 | 91.30 | 85.71 | 77.78   | 64.71   | 84.33 |
| Gemma-2B | ReAct     | 0%     | 0.00   | 9.68  | 0.00  | 4.76  | 44.44   | 0.00    | 8.96  |
| Gemma-2B | ETO       | 0%     | 91.67  | 83.87 | 78.26 | 52.38 | 77.78   | 29.41   | 71.64 |
| Gemma-2B | KnowAgent | 100%   | 91.67  | 90.32 | 69.57 | 71.43 | 66.67   | 41.18   | 73.88 |
| Gemma-2B | WKM       | 100%   | 91.67  | 87.10 | 78.26 | 71.43 | 61.11   | 52.94   | 76.12 |
| Gemma-2B | KnowSelf  | 26.41% | 87.50  | 93.55 | 73.91 | 76.19 | 83.33   | 52.94   | 79.85 |

# Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



**Efficiency signal**

- KnowSelf (Llama-8B): 84.33% with only 15.01% knowledge usage.
- KnowSelf (Gemma-2B): 79.85% with 26.41% knowledge usage.
- ExpeL / KnowAgent / WKM rely on 100% knowledge yet often score lower.

**Scaling insight**

Selective retrieval improves both small and mid-size backbones.

# Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- 1) Indiscriminate knowledge injection is wasteful and can degrade planning quality.
- 2) Situational self-awareness is learnable with lightweight heuristic labeling and special tokens.
- 3) KnowSelf (Llama-8B) outperforms strong baselines: 84.33% vs GPT-4o ReAct 76.12 and GPT-4o ExpeL 79.85.
- 4) Knowledge usage drops to 15-26% versus 100% in ExpeL/KnowAgent/WKM.
- 5) Paradigm transfers across scales: Gemma-2B + KnowSelf reaches 79.85%.

Future directions: richer self-awareness labels, broader environments beyond ALFWorld, and joint optimization of retrieval policy + planner.